



Dhriti Kotwal

Nationality: Indian **Date of birth:** 30/09/2005 **Gender:** Female

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Website: <https://dhritikotwal.github.io/>

Work: UM-DAE Centre for Excellence in Basic Sciences, Mumbai University, Kalina Campus, Santacruz East, 400098 Mumbai (India)

ABOUT ME

Undergraduate BS-MS student focused on cosmology, with interests in the cosmic microwave background, Type Ia supernovae, and questions in precision cosmology including cosmic acceleration and current observational tensions. Active in science outreach and institute-level initiatives.

EDUCATION AND TRAINING

BS-MS Integrated Program - Physics

UM-DAE Centre for Excellence in Basic Sciences [2023 – 2028]

City: Mumbai | Website: <https://www.cbs.ac.in/>

WORK EXPERIENCE

Dr. Suvedha Naik – Indian Institute of Astrophysics, Bangalore

Summer Intern

[15/05/2025 – 15/07/2025]

Conducted a research project at the IIA, Bangalore, using Planck CMB data to study the Universe's energy budget. Simulated CMB angular power spectra with the CLASS code for varying cosmological parameters and obtained the best-fit Λ CDM model by comparing with observational data.

Prof. Sathish Akella – Indian Institute of Technology, Jammu

Summer Intern

[06/2024 – 07/2024]

MATLAB simulation-based research project at IIT, Jammu on statistical physics, focusing on Brownian motion and Langevin equations.

Dr. Bhooshan Paradkar – UM-DAE CEBS, Mumbai

Reading Project

[01/2026 – Current]

Reading project on General Relativity, studying core concepts using Prof. David Tong's lecture notes.

Dr. Ameeya Bhagwat, Dr. Rajesh Vatsa – UM-DAE CEBS, Mumbai

Documentary - Astronomy in Daily Lives

[03/2026 – Current]

Documentary project exploring how astronomical research drives technological development, featuring discussions with prominent researchers.

SKILLS

Python / Fortran / MATLAB / CLASS (Cosmological Linear Anisotropy Solving System) / Linux / HTML / LaTeX

HONOURS AND AWARDS

Department of Atomic Energy, India

DISHA Scholarship

[2026] Graduate Aptitude Test in Engineering (GATE)

AIR 510

[2026] Joint Admission Test for Masters (IIT JAM)

AIR 286

[2023] National Entrance Screening Test (NEST)

AIR 473, State (JK) Rank 2

SOCIAL AND POLITICAL ACTIVITIES

[02/2025 – Current]

Active Member of Breakthrough Science Society (BSS)

Links: <https://breakthroughindia.org/> | <https://www.youtube.com/@breakthroughsciencesociety306>

[2024 – 2025]

Science Week Volunteers at UM-DAE CEBS

CREATIVE WORKS

[01/2026 – Current]

Cultural Secretary, UM-DAE CEBS

Link: <https://www.youtube.com/@UMDAECEBSCultural>

[01/2024 – Current]

Coordinator, Cinematography Club, UM-DAE CEBS

Link: <https://www.youtube.com/@cebsfilmmakingclub>